

3 CAD Laboratory Session n°3 - Design of a capacitive feedback amplifier based on a two-stage OTA

3.1 OTA schematic and testbenches used

This laboratory session focuses on the analysis and design of a capacitive feedback amplifier based on a two-stage Operational Transconductance Amplifier (OTA). Initially, the transistor dimensions were provided without a compensation network. Fig. 25 illustrates the core circuit schematic, while Fig. 26 and Fig. 27 present the open-loop and closed-loop test benches used to evaluate the system's performance.

The target specifications for the feedback amplifier are outlined below:

- Static error: $\epsilon_0 \leq 0.2\%$
- Dynamic error: $\epsilon_d \leq 0.2\%$ at a settling time $t_s = 20$ ns
- Phase Margin: $PM \leq 70^\circ$
- Ideal low-frequency closed-loop gain: $A_\infty = -2$
- OTA DC power consumption: $P_{DC} \leq 1.2$ mW
- Supply Voltage: $V_{DD} = 1.2$ V
- OTA offset voltage: $V_{OS} \leq 2$ mV
- External load capacitance: $C_L = 4$ pF
- Source capacitance: $C_S = 1$ pF
- Reference current: $I_{REF} = 50$ μ A

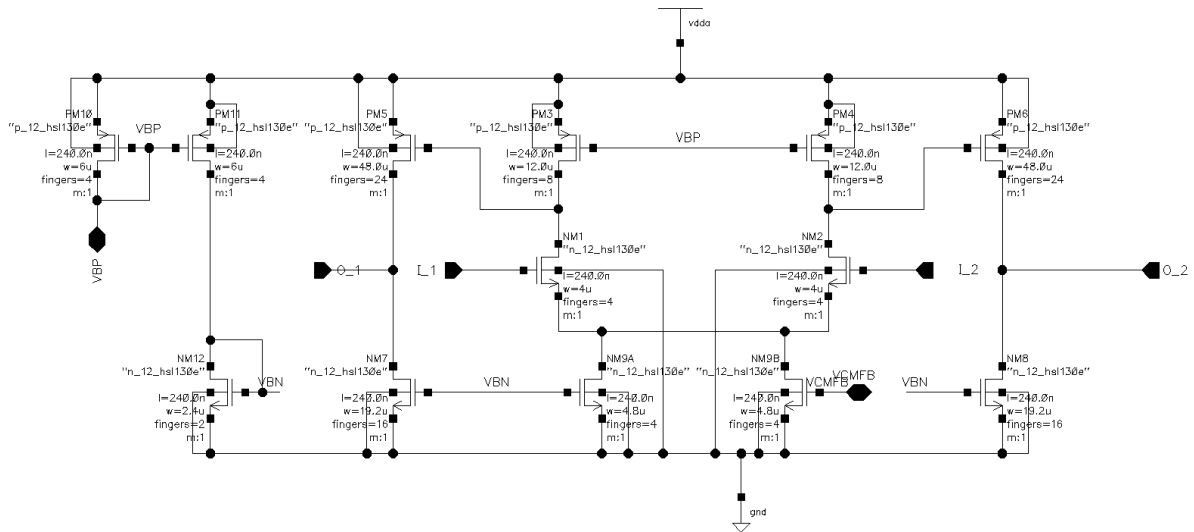


Figure 25: Schematic Initial Design

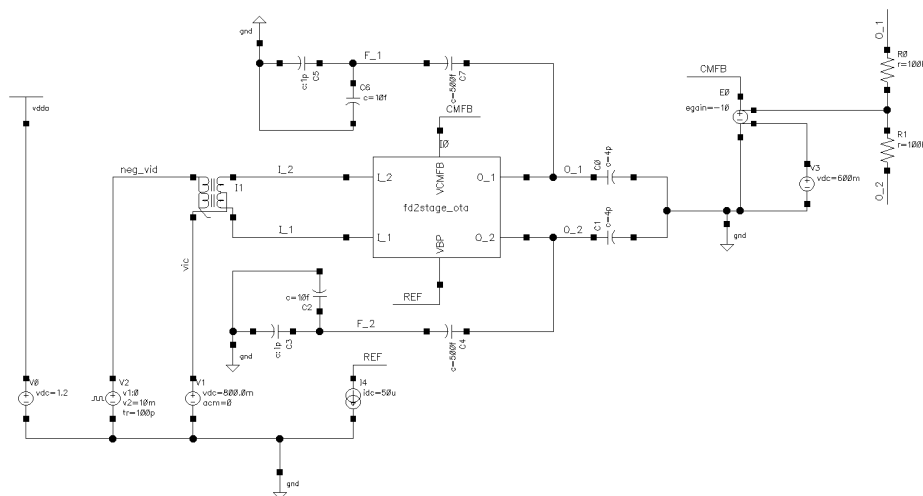


Figure 26: Testbench Open Loop

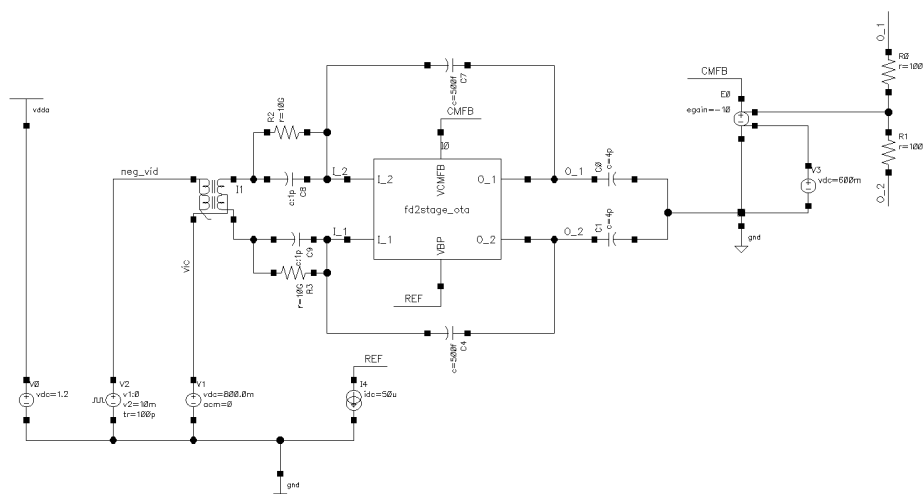


Figure 27: Testbench Closed Loop

3.2 DC, AC and Transient Simulations

To analyze the open-loop and closed-loop behavior, three distinct simulations were performed. These aimed to determine the transistor operating regions by evaluating their terminal voltages, assess the differential component at the OTA output port, and measure the returning differential signal.

The operating points obtained from the DC simulation are summarized in the table above, confirming that all transistors are biased within the saturation region.

First, it was done a simulation using the open-loop testbench to see the bode plot (see Fig. 28) and the transient response (see Fig. 29) of the loop transfer function.

Transistor	$ I_d [\mu A]$	$ V_{gs} [mV]$	$ V_{ds} [mV]$	$ V_{th} [mV]$	$ V_{ov} [mV]$	g_m	$r_o[k\Omega]$
$M_1 - M_2$	100	493.779	398.908	353.313	139.966	1.06426 mS	26.5827
$M_3 - M_4$	100	489.337	494.871	281.973	207.364	841.767 μS	36.8114
$M_5 - M_6$	444.188	494.871	644.434	279.155	215.716	3.5886 mS	8.84757
$M_7 - M_8$	444.188	446.599	555.566	316.792	129.802	5.03376 mS	7.10217
M_{9A}	101.456	446.599	306.221	317.186	129.413	1.16647 mS	22.0618
M_{9B}	98.8432	444.344	306.221	317.186	127.158	1.15138 mS	22.631
M_{10}	50	489.337	489.337	282.046	207.291	420.398 μS	73.51
M_{11}	53.558	489.337	753.401	278.576	210.761	442.038 μS	73.8276
M_{12}	53.558	446.599	446.599	316.964	129.635	611.753 μS	53.7231

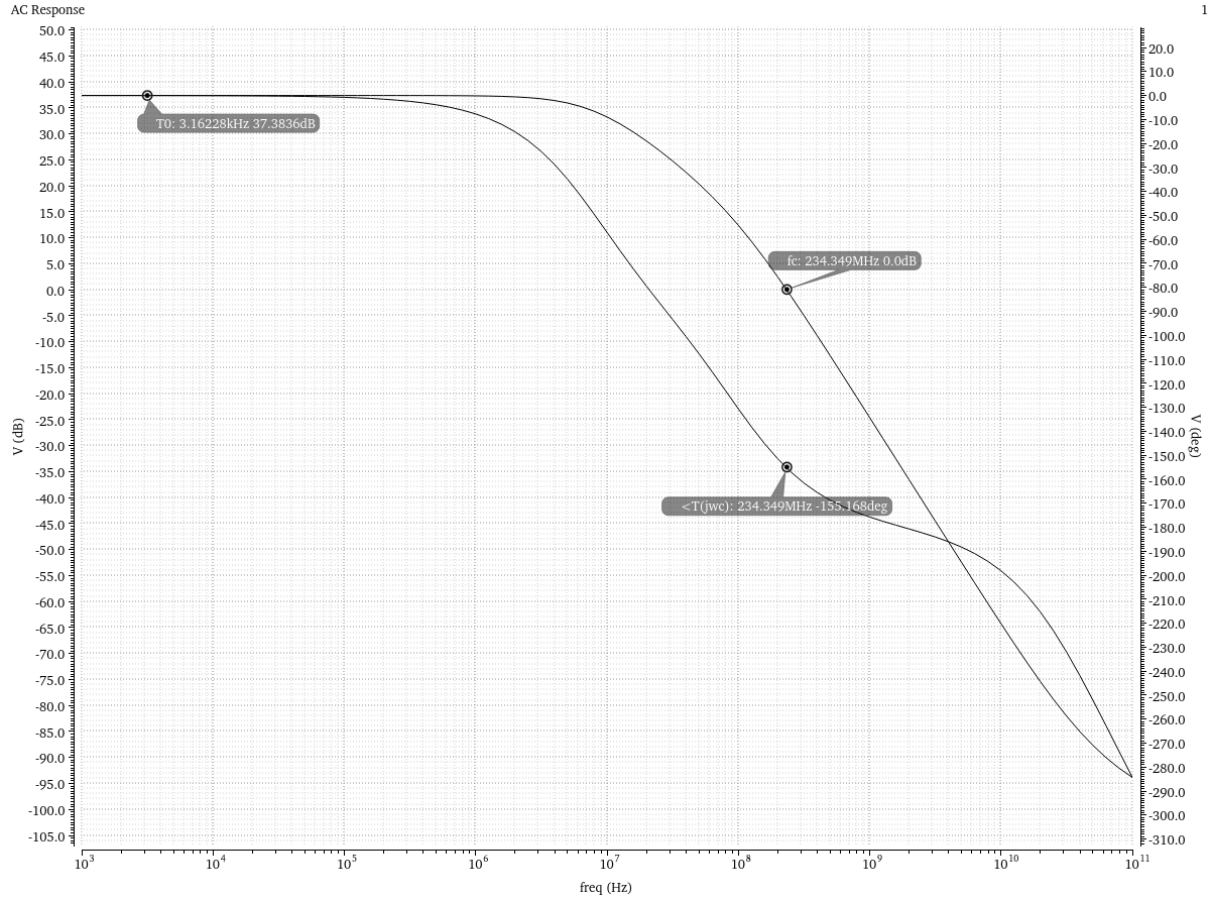


Figure 28: Bode Plot - Open Loop

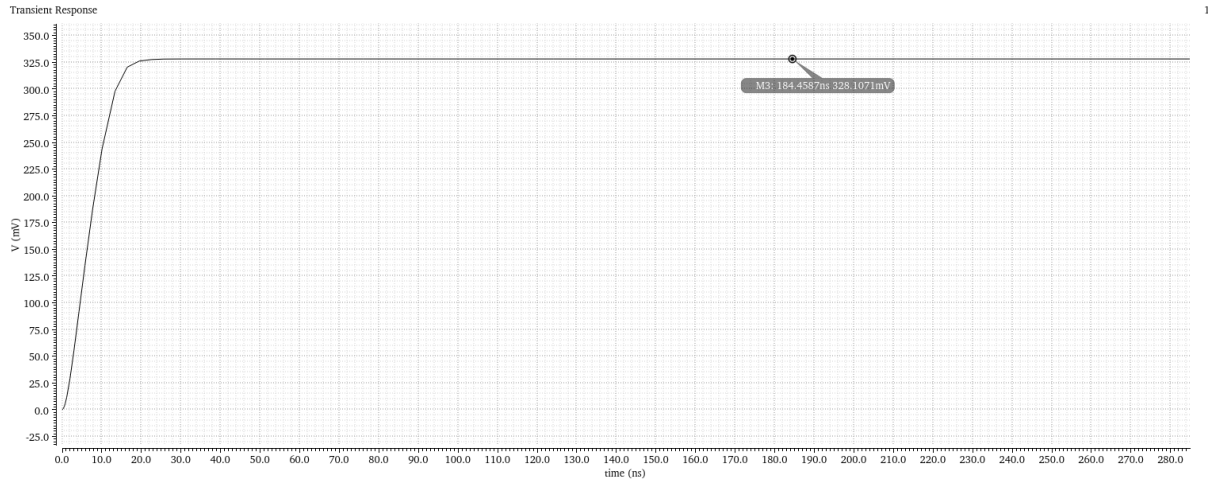


Figure 29: Transient in Open Loop

Thanks to the AC simulation done, it is possible to evaluate the crossover frequency, $f_c = 234.349 \text{ MHz}$, and see how the phase margin with the values given is $PM = 180 - 155.168 = 24.832$, which is very low. It is possible also to see the behavior at low frequency and see the value of the magnitude to be 37.3836 dB .

While, in the transient simulation it possible to say that, since there is no feedback, the curve is smooth and has no overshoot. After this, the closed-loop testbench was used to evaluate the behavior of the feedback amplifier transfer function (see Fig. 30) and transient response (see Fig. 31)

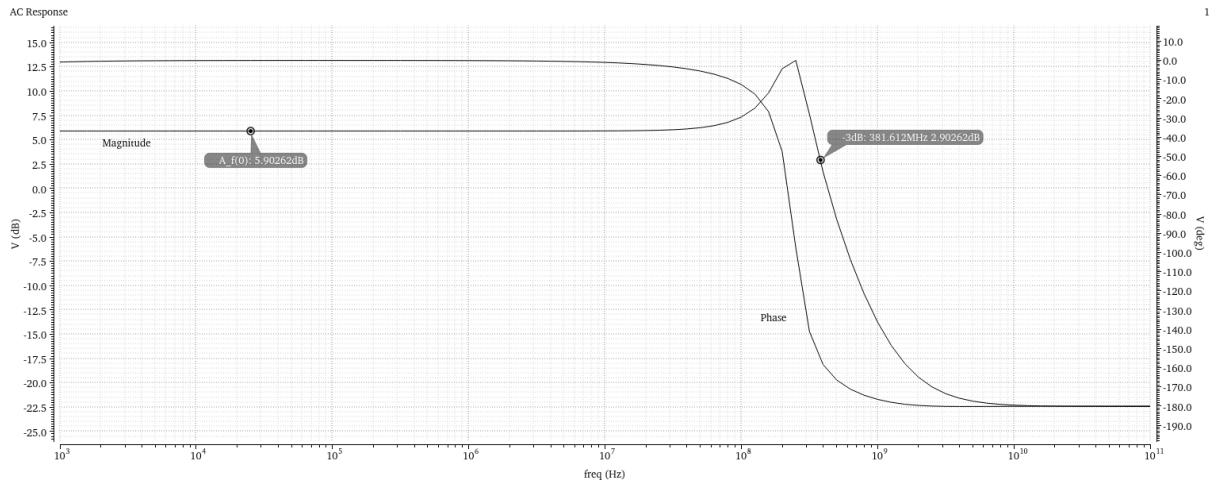


Figure 30: Bode Plot - Closed Loop

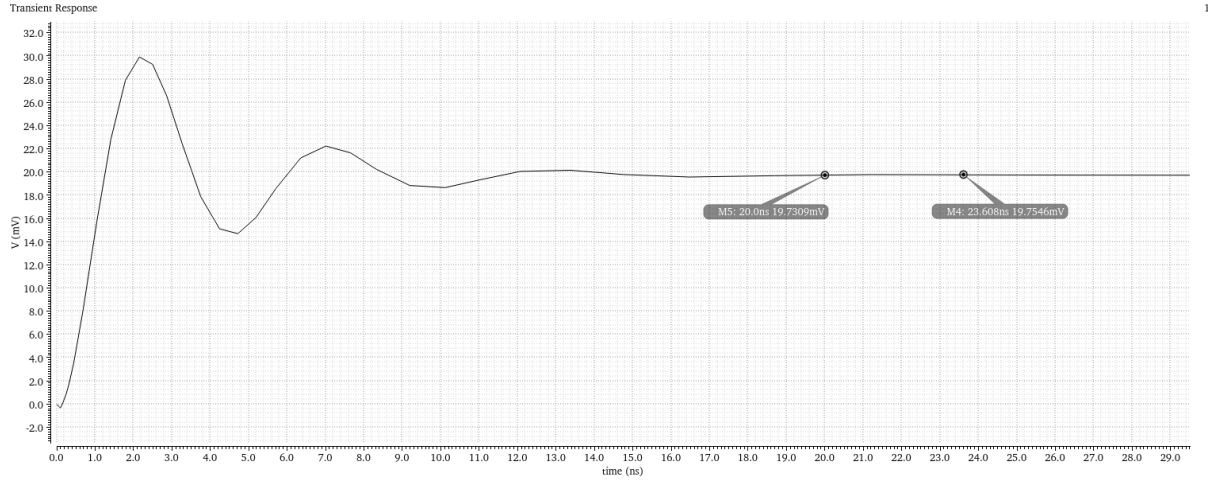


Figure 31: Transient in Closed Loop

Closing the feedback drops the low frequency gain to 5.90262 dB , The system's -3 dB bandwidth expands to 381.612 MHz . It is possible also to notice the bulge the magnitude has, peaking up to 13 dB , which is the consequence of the 24.8 open-loop phase margin. The transient response also shows some instability: severe ringing and overshoot before settling to its final value, which is 19.7546 mV . This behavior is also due to the low phase margin analyzed before and the bulge in the magnitude of the closed-loop. Thanks to Fig. 32 it's also possible noticing how the offset voltage does not meet the specification.

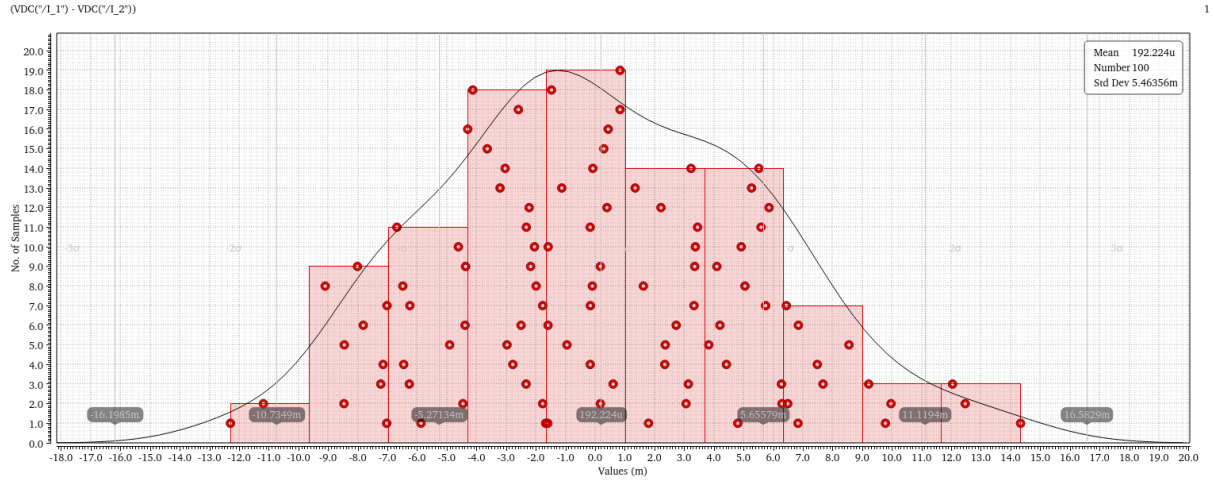


Figure 32: Initial Monte Carlo Simulation

3.3 Design of the two-stage OTA

Since it was possible to notice how the specifics were not met, it was have to be made the design of the two-stage OTA from scratch, starting from the specifics that they were given. It was possible to derive the value of the capacitance C_f from the spec 4 :

$$A_{\infty} = -\frac{C_s}{C_f} \Rightarrow C_f = -\frac{C_s}{A_{\infty}} = -\frac{1 \text{ pF}}{-2} = 0.5 \text{ pF}$$

Assuming the OTA parasitic input capacitance C_i negligible, it was possible to compute the parameter β :

$$\beta = \frac{C_f}{C_f + C_i + C_s} \approx \frac{C_f}{C_f + C_s} = \frac{1}{3}$$

From spec 1, thanks to the static error it was possible to derive the value of the loop gain T_0 and then the OTA low-frequency gain A_{dd0} :

$$\epsilon_0 = \frac{1}{1 + T_0} \implies T_0 = \frac{1 - \epsilon_0}{\epsilon_0} = 499 \approx 500$$

$$A_{dd0} = \frac{T_0}{\beta} = 1500$$

To derive the loop transfer function crossover frequency $\omega_C = 2\pi f_C$ and the OTA GBW, it was used the spec 2 :

$$\epsilon_d \approx e^{-\omega_C t_s} \implies \omega_C = \frac{1}{t_s} \ln(1/\epsilon_d) = 310.73 \text{ Mrad/s} = 2\pi(49.45 \text{ MHz})$$

$$\omega_C = \beta \cdot \omega_u \implies \omega_u = \frac{\omega_C}{\beta} = 931.19 \text{ Mrad/s} = 2\pi f_u = 2\pi(148.36 \text{ MHz})$$

Supposing initially the compensating capacitance $C_C = \frac{C_L}{2} = 2 \text{ pF}$, it was possible to derive the transconductance G_{m1} of the first stage :

$$G_{m1} = \frac{\omega_C \cdot C_C}{\beta} = 1.8644 \text{ mS}$$

Assuming an overdrive voltage for the differential pair of $V_{OV1} = 0.15 \text{ V}$ and having the transconductance G_{m1} ,

$$\text{Differential pair bias current } I_{B1}: I_{D1} = \frac{V_{OV1} \cdot G_{m1}}{2} \approx 140 \text{ }\mu\text{A} \implies I_{B1} = 2 \cdot I_{D1} = 280 \text{ }\mu\text{A}$$

Using spec 5 and 6, it was possible to derive the maximum bias current I_{B2_MAX} of each of the two branches of the second stage :

$$\frac{P_{DC}}{V_{DD}} = 1000 \text{ }\mu\text{A} = I_{B1} + 2I_{B2_MAX}$$

$$I_{B2_MAX} = 360 \text{ }\mu\text{A}$$

Assuming the overdrive voltage of the second stage $|V_{OV5}| = 0.2 \text{ V}$ and choosing $I_{B2} = 300 \text{ }\mu\text{A}$:

$$\text{Transconductance of the second stage : } G_{m2} = \frac{2I_{B2}}{|V_{OV5}|} = 3 \text{ mS}$$

Imposing a phase margin of 70, it was possible to evaluate the poles and zeros of the transfer function and find a value for the compensating resistance :

$$PM = 70 = 180 - 89.9 - tg^{-1}\left(\frac{\omega_u}{\omega_{p2}}\right) + \phi_z \quad \text{where :} \quad \omega_{p2} = \frac{G_{m2}}{C_L}$$

$$\phi_{p2} = 22.5 \implies \phi_z = 2.5 < 10$$

Since the zero is lower than 10, it is needed to increase. To do that, the solution used was to increase the current along the differential pair, so to increase also the transconductance and so on. So I_{D1} was set to $I_{D1} = 200 \mu A$, so the new parameter values will be :

$$I_{B1} = 400 \mu A \quad G_{m1} = \frac{I_{B1}}{V_{OV1}} = 2.667 mS \quad I_{B2_MAX} = 300 \mu A$$

Choosing $I_{B2} = 250 \mu A$ and setting $|V_{OV5}| = 0.15 V$:

$$G_{m2} = 3.33 mS \quad \phi_{p2} = 28.07.$$

By setting also the phase margin to 80, $\phi_z = 18.07$. It is now possible to evaluate the compensating resistance :

$$\omega_z = \frac{\omega_u}{tg(\phi_z)} = 2\pi(454.7 MHz) = \frac{1}{C_C(R_Z - \frac{1}{G_{m2}})} \implies R_Z = \frac{1}{G_{m2}} + \frac{1}{C_C\omega_z} \approx 500 \Omega$$

The new value of the OTA low-frequency gain A_{dd0} will be :

$$A_{dd0} = G_{m1}R_1G_{m2}R_2 \quad \text{where} \quad R_1 = r_{o1}/r_{o3} \quad R_2 = r_{o5}/r_{o7} \quad r_o = \frac{\eta L}{I_D}$$

By setting $L_1 = L_2 = L_5 = L_6 = 480 nm$ and $L_3 = L_4 = L_7 = L_8 = 600 nm$:

$$r_{o1} = 26.4 k\Omega \quad r_{o3} = 45 k\Omega \quad r_{o5} = 28.8 k\Omega \quad r_{o7} = 26.4 k\Omega$$

$$R_1 = 16.64 k\Omega \quad R_2 = 13.77 k\Omega$$

$$A_{dd0} = G_{m1}R_1G_{m2}R_2 = 2.666 mS \cdot 16.64 k\Omega \cdot 3.33 mS \cdot 13.77 k\Omega \approx 2037 (66.2 dB)$$

Since it is verified that $A_{dd0} > 1500$, it is possible now to evaluate the gate width of all transistors, imposing the overdrive voltage for M_k , where $k = 3, 4, 7, 8, 9, 10, 11, 12$, equal to $|V_{OVk}| = 0.3 V$

$$W_1 = W_2 = \frac{2I_{D1}L_1}{k'V_{OV1}^2} = 18.16 \mu m$$

$$W_3 = W_4 = 26.67 \mu m \rightarrow 26.68 \mu m$$

$$W_5 = W_6 = 106.67 \mu m \rightarrow 106.68 \mu m$$

$$W_7 = W_8 = 7.09 \mu m$$

Assuming the offset voltage V_{OS} it is entirely due to the first stage :

$$\sigma_{\Delta V_{Tn}} = \frac{A_{\Delta V_{Tn}}}{\sqrt{W_1 \cdot L_1}} = 1.694 mV \quad \sigma_{\frac{\Delta k'_n}{k'_n}} = \frac{A_{k'_n}}{\sqrt{W_1 \cdot L_1}} = 0.373\%$$

$$\sigma_{\Delta V_{Tp}} = \frac{A_{\Delta V_{Tp}}}{\sqrt{W_3 \cdot L_3}} = 0.9 mV \quad \sigma_{\frac{\Delta k'_p}{k'_p}} = \frac{A_{k'_p}}{\sqrt{W_3 \cdot L_3}} = 0.25\%$$

$$V_{OS} = \sigma_{V_{ID}} = \sqrt{\left(\frac{V_{OV1}}{2}\right)^2 \left(\sigma_{\frac{\Delta k'_p}{k'_p}}^2 + \sigma_{\frac{\Delta k'_n}{k'_n}}^2\right) + \left(\frac{V_{OV1}}{V_{OV3}}\right)^2 \sigma_{\Delta V_{Tp}}^2 + \sigma_{\Delta V_{Tn}}^2} = 1.78 mV$$

3.4 Simulation after the design of the capacitive feedback amplifier based on a two-stage OTA

After the design, the same simulations done before were repeated. The value of the compensating resistance was incremented to $R_C = 1.1k\Omega$ in order to satisfy the phase margin specific. In Fig. 33 it's possible to notice that all transistors operate in the saturation region.

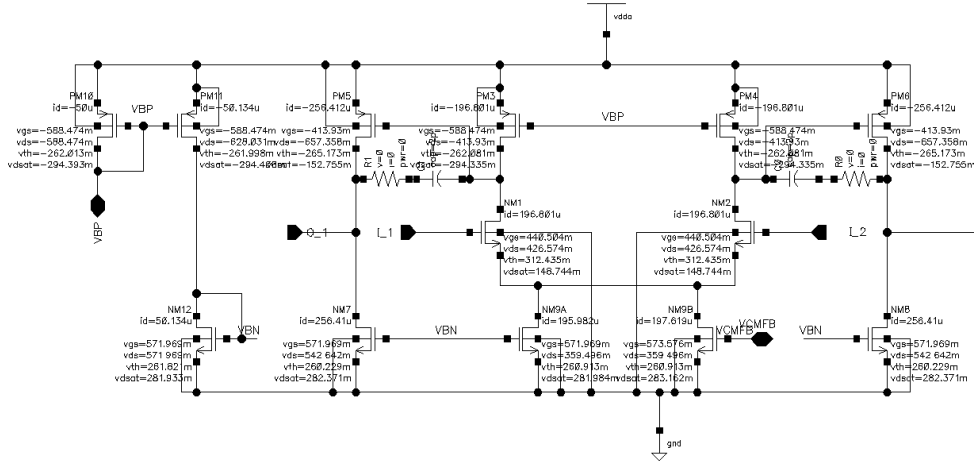


Figure 33: DC Operating Points

Then, as before, the simulation were done to see the new behavior of the circuit.

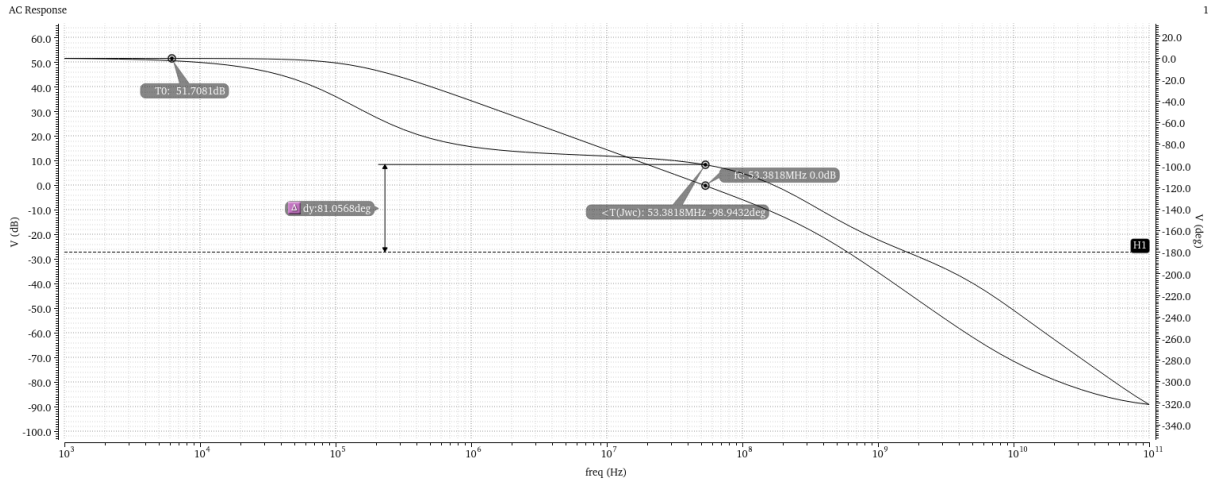


Figure 34: Bode Plot in Open Loop after the design

The AC simulation shown in Fig. 34, shows how, after the design, the gain at low frequency $T_0 = 51.7081 \text{ dB}$, the crossover frequency is almost the same as the one found theoretically, $f_C = 53.3818 \text{ MHz}$, having a phase margin of ≈ 81 , meeting the specification. While, in the AC simulation in closed loop (see Fig. 35) it shows the low frequency gain $\approx 6 \text{ db}$.

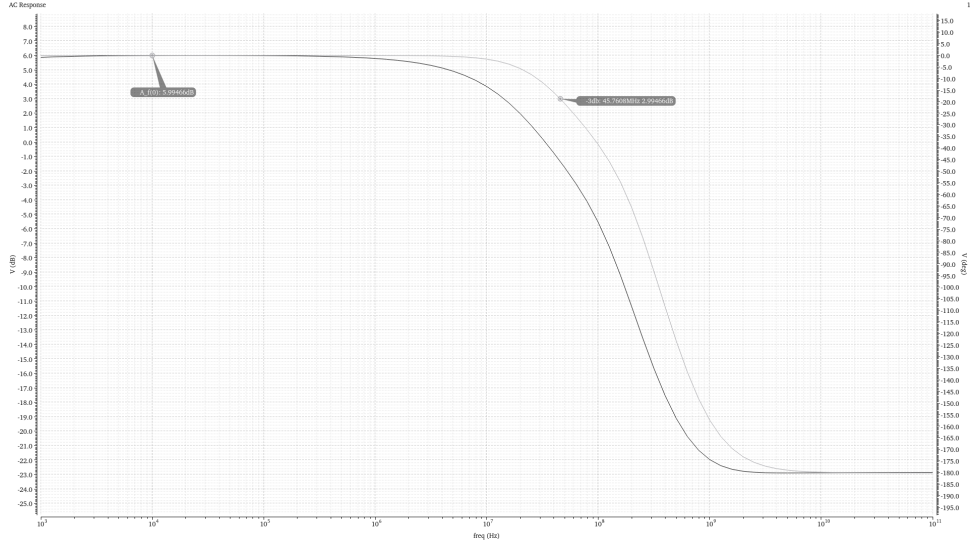


Figure 35: Bode Plot in Closed Loop after the design

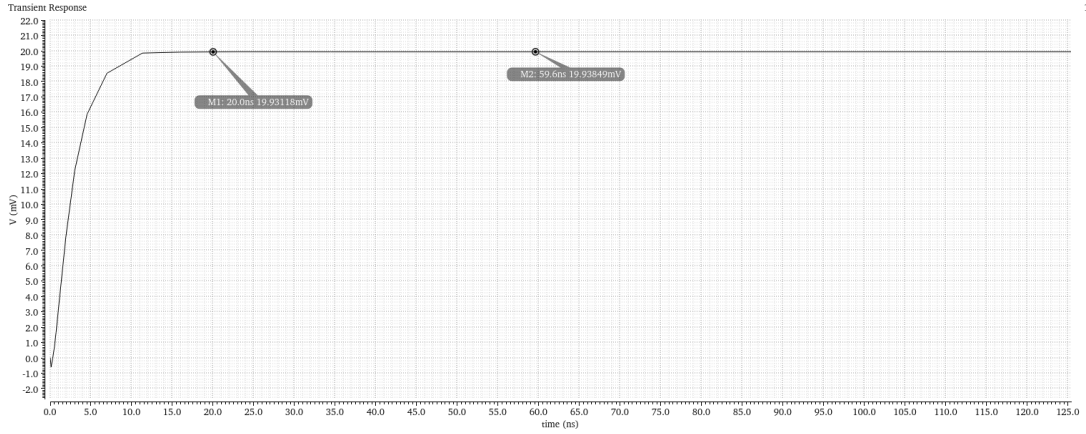


Figure 36: Transient in Closed Loop

In the end, it is also possible to notice the new transient curve (see Fig. 36) that shows how the overshoot disappears and the system is much more stable. It is also possible to find the values of the dynamic error and static error :

$$\text{Relative Static Error : } \epsilon_o = \left| \frac{v_o^{id}(\infty) - v_o(\infty)}{v_o^{id}(\infty)} \right| = 0.3\%$$

$$\text{Relative Dynamic Error : } \epsilon_d = \left| \frac{v_o(ts) - v_o(\infty)}{v_o^{id}(\infty)} \right| = 0.037\%$$

3.4.1 Monte Carlo Simulation

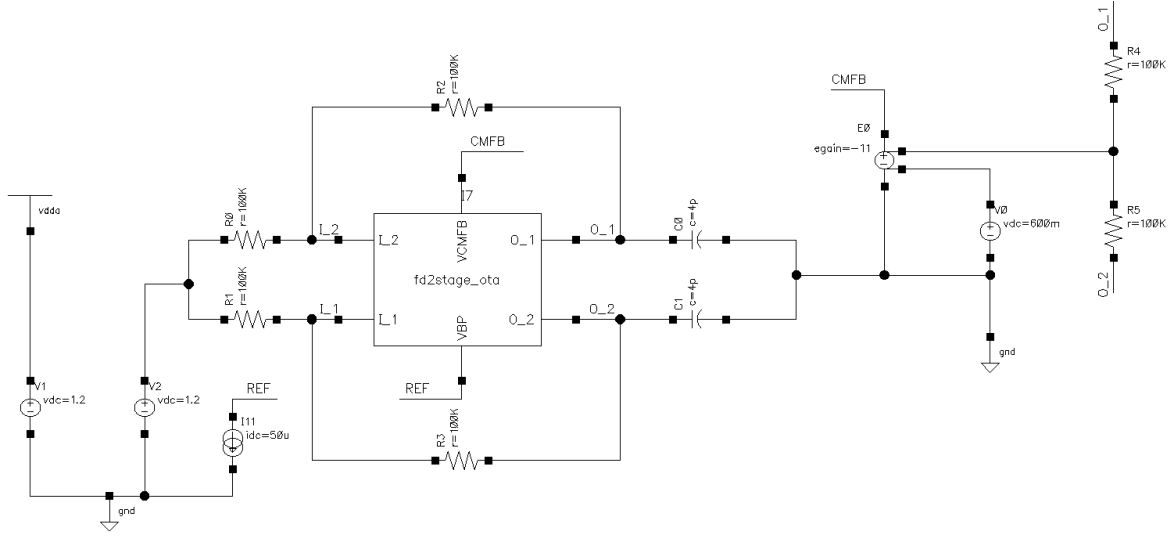


Figure 37: Test bench used for the Monte Carlo simulation

Figure 37 illustrates the test bench implemented for the Monte Carlo analysis. The simulation results, presented in Fig. 38, confirm that the offset voltage specification is successfully satisfied, yielding a calculated standard deviation of $\sigma = 1.61 \text{ mV}$, which is well within the 2 mV maximum limit.

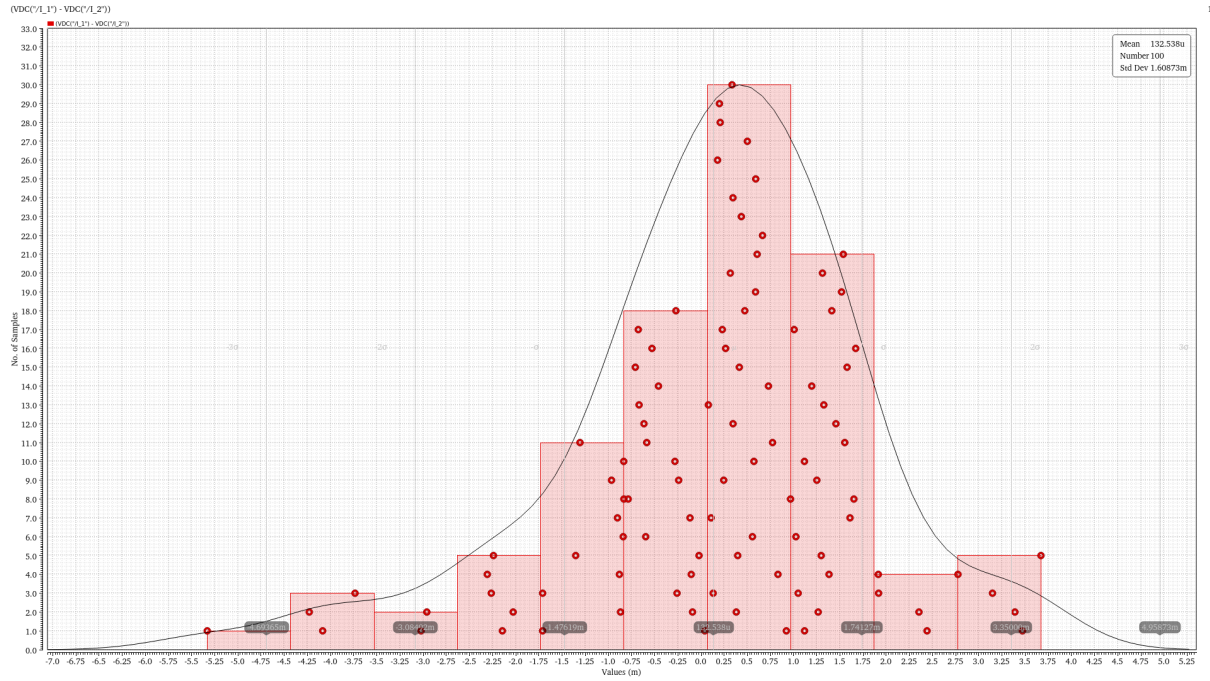


Figure 38: Monte Carlo Simulation